Aim

To implement a **Feed Forward Neural Network (FFNN)** using Python for classification.

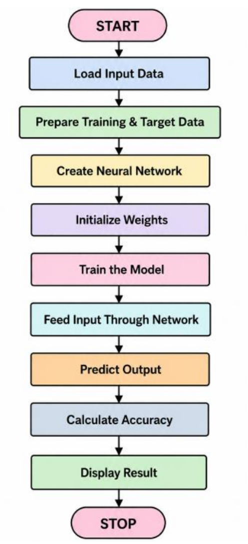
Objectives

- To understand the basic structure of a neural network.
- To implement input, hidden, and output layers.
- To train the neural network using sample data.
- To predict the output for new input data.

Algorithm

1. Start.
2. Prepare the input and target data.
3. Create the Feed Forward Neural Network.
4. Initialize the network weights.
5. Train the model using the training data.
6. Pass the input through the network.
7. Calculate the predicted output.
8. Evaluate the model accuracy.
9. Display the prediction and accuracy.
10. Stop.

Flowchart



Program

```python
from sklearn.neural_network import MLPClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1],
    [0, 0],
    [1, 1]
]

y = [0, 1, 1, 0, 0, 0]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=42
)

model = MLPClassifier(
    hidden_layer_sizes=(4,),
    activation='relu',
    max_iter=2000,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Predicted Output:", y_pred)
print("Accuracy:", accuracy)
```

Output

```
Predicted Output: [0 0]
Accuracy: 0.5
>>>
```

Result

The **Feed Forward Neural Network** was successfully implemented using Python, and the model predicted the test data with the calculated accuracy.